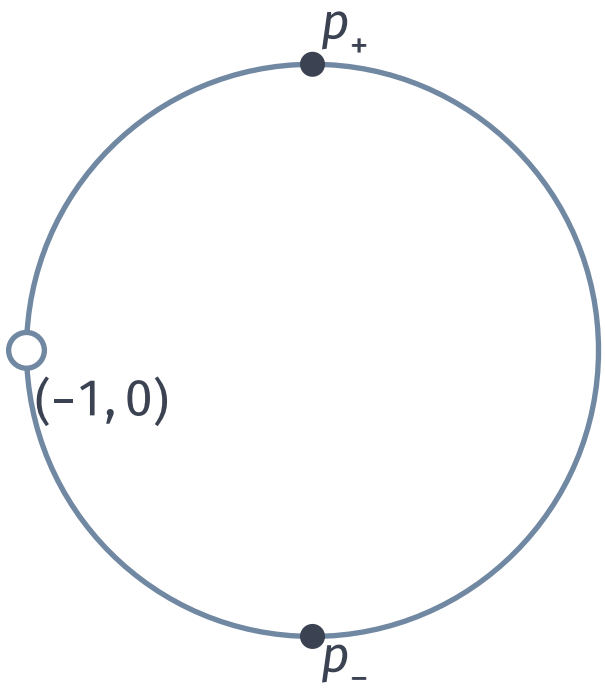
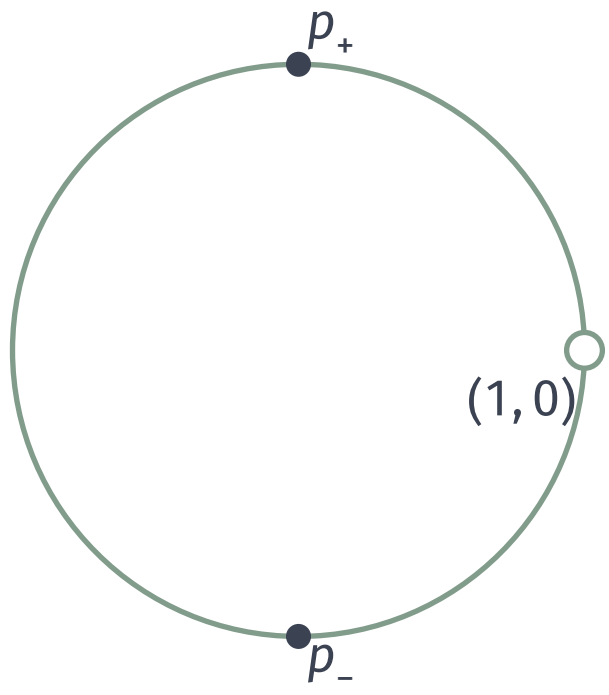
$\chi_1 : \ \theta \in (-\pi, \pi)$

$\chi_2 : \ \theta' \in (0, 2\pi)$





$p_+ : \ \theta = \pi/2 \longrightarrow p_+ : \ \theta' = \pi/2$

$p_- : \ \theta = -\pi/2 \longrightarrow p_- : \ \theta' = 3\pi/2$

$\theta > 0 : \ \theta' = \theta; \qquad \theta < 0 : \ \theta' = \theta + 2\pi$